THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The length of the arc of a sector of angle 90° in a circle of radius 14 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 24

2. A triangle has area 84 cm^2 and one side 14 cm. The altitude on this side (cm) is: 1

(a) 13 **(b)** 14 **(c)** 12 **(d)** 11

3. The height of a parallelogram of area 204 cm^2 and base 17 cm is (cm): 1

(a) 11 **(b)** 14 **(c)** 13 **(d)** 12

4. A square has area 169 cm^2. Its perimeter (cm) is: 1

(a) 51 **(b)** 53 **(c)** 54 **(d)** 52

5. The area of a quadrilateral whose diagonal is 20 cm and whose offsets from the diagonal are 6 cm and 8 cm is (cm^2): 1

(a) 139 **(b)** 141 **(c)** 142 **(d)** 140

6. A rectangle of perimeter 50 cm has length 18 cm. Its area (cm^2) is: 1

(a) 126 **(b)** 125 **(c)** 127 **(d)** 128

7. A circular wire of radius 7 cm (take $\pi = \frac{22}{7}$) is bent into a square. The side of the square (cm) is: 1

(a) 11 **(b)** 12 **(c)** 10 **(d)** 13

8. A wheel of diameter 70 cm (take $\pi = \frac{22}{7}$) makes 50 revolutions. The distance covered (in metres) is: 1

(a) 112 **(b)** 111 **(c)** 110 **(d)** 109

9. The length of the arc of a sector of angle 240° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 90 **(b)** 89 **(c)** 88 **(d)** 87

10. A triangle has area 84 cm^2 and one side 21 cm. The altitude on this side (cm) is: 1

(a) 9 **(b)** 7 **(c)** 10 **(d)** 8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A triangle has sides 25 cm, 25 cm and 14 cm. Find its area and the altitude on the side 14 cm. 2

12. Find the area of a quadrilateral $ABCD$ in which diagonal $AC = 26$ cm and the perpendiculars from B and D to AC are 9 cm and 7 cm. 2

13. Find the arc length for 40° in a circle of radius 63 cm (take $\pi = \frac{22}{7}$). 2

14. A triangle has sides 13 cm, 20 cm and 21 cm. Find its area and the altitude on the side 21 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the cyclic quadrilateral with sides 4 cm, 8 cm, 11 cm and 17 cm. 3

16. A rectangular sheet 30 cm $\times$ 14 cm has a semicircle of diameter 14 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

17. The sides of a parallelogram are 15 cm and 12 cm and the height on the side 15 cm is 8 cm. Find its area and the height on the side 12 cm. 3

OR

Find the area of the cyclic quadrilateral with sides 8 cm, 9 cm, 12 cm and 19 cm. 3

18. A rectangular sheet 25 cm $\times$ 28 cm has a semicircle of diameter 28 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

OR

The sides of a parallelogram are 18 cm and 9 cm and the height on the side 18 cm is 5 cm. Find its area and the height on the side 9 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular plot 40 m $\times$ 30 m has a circular pond of radius 7 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹6 per m^2. 5

OR

A triangular field has sides 13 m, 13 m and 10 m. Find its area, and the cost of ploughing it at ₹5 per m^2. 5

20. A rectangular plot 50 m $\times$ 40 m has a circular pond of radius 14 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹4 per m^2. 5

OR

A triangular field has sides 15 m, 15 m and 24 m. Find its area, and the cost of ploughing it at ₹4 per m^2. 5

END OF QUESTION PAPER — Total: 40 Marks

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